

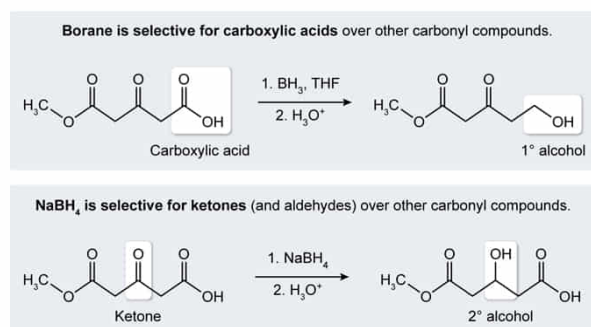
Two separate reactions are conducted in which a compound containing a ketone, an ester, and a carboxylic acid is reacted with borane (BH_3) in THF in one reaction and with NaBH_4 in methanol in the other. Which of the following explains why different products are observed?

- ☐ A. BH_3 will selectively reduce ketones, and NaBH_4 will only reduce carboxylic acids. [23%]
- ☐ B. BH_3 does not reduce ketones, and NaBH_4 will selectively reduce esters. [12%]
- ☐ C. BH_3 reduces carboxylic acids and esters, and NaBH_4 will only reduce esters. [14%]
- ✔ ☒ D. BH_3 will selectively reduce carboxylic acids, and NaBH_4 will selectively reduce ketones. [49%]

Correct

49% answered correctly

Explanation:



Reduction of a functional group results in a gain of electrons from the transfer of a hydride, a decreased **oxidation state** of carbon, and a decreased number of C–O bonds. Several reagents can be used to reduce a functional group, including borane (BH_3) and NaBH_4 . These **reducing agents** transfer one or more hydrides to the carbon atom being reduced. BH_3 reacts most readily with carboxylic acid carbonyl groups and therefore will **selectively reduce carboxylic acids**. As a result, the intermediate aldehyde will be further reduced to a primary alcohol. NaBH_4 reduces the **reactive carbonyls**—ketones and aldehydes—and **will not reduce** the less reactive **esters** and **carboxylic acids**.

If a compound that contains a ketone, an ester, and a carboxylic acid is reacted with BH_3 , the carboxylic acid will be selectively reduced to a primary alcohol, leaving the ketone and ester intact. When the same compound is reacted with NaBH_4 , only the

ketone will be reduced, generating a secondary alcohol. Therefore, the products of the two reactions will be different.

(Choice A) BH_3 and NaBH_4 are both reducing agents, but BH_3 selectively reduces carboxylic acids rather than ketones, and NaBH_4 is not strong enough to reduce carboxylic acids. NaBH_4 selectively reduces ketones and aldehydes.

(Choice B) Although it is true that BH_3 does not reduce ketones, NaBH_4 *does not* reduce esters.

(Choice C) Esters cannot be reduced by BH_3 or NaBH_4 ; this functional group can only be reduced to a primary alcohol by LiAlH_4 .

Educational objective:

Carboxylic acids can be selectively reduced to primary alcohols by the reducing agent BH_3 . They cannot be reduced by NaBH_4 , which is selective for the reactive carbonyls ketones and aldehydes.

Time spent:0 sec

Subject:

Organic Chemistry

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System:

Functional Groups and Their Reactions